NETWORKING PROJECT REPORT

Design and Implementation of a Site-to-Site IPsec VPN Between Two Corporate Branch Offices

New York (192.168.10.0/24) and London (192.168.20.0/24)

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Simulation Tool	Cisco Packet Tracer (Cisco 2911 routers, Cisco 2960 switches)

1. Abstract

This project implements a site-to-site IPsec Virtual Private Network (VPN) between two simulated corporate branch offices, New York and London. Each office has its own LAN with five PCs connected through a switch to an edge router. The two routers are joined by a simulated WAN link. Static routes let each router reach the remote LAN, and an IKEv1/IPsec VPN encrypts the traffic between the two networks. The implementation was built and verified in Cisco Packet Tracer using ping tests together with the router commands *show crypto isakmp sa* and *show crypto ipsec sa*.

2. Introduction and Problem Statement

A company with offices in New York and London needs the two sites to behave as one network, so that staff can reach servers and PCs in the other city. The only path between the sites is an untrusted public network, where traffic can be read or modified in transit. A site-to-site VPN solves this: the edge routers build an encrypted tunnel and protect all traffic that passes between the two private networks, while users carry on using ordinary IP addresses.

The two offices use different IP subnets and are joined by routing, not by one shared Ethernet segment. Routing decides where a packet goes. IPsec decides how it is protected on the way.

3. Objectives

- Build two independent office LANs, each with five PCs, a switch and an edge router.
- Configure router LAN and WAN interfaces and assign addresses and gateways to all ten PCs.
- Configure static routes so each router knows the remote LAN is reached through the other router.
- Configure an IKEv1/IPsec site-to-site VPN between the two edge routers.
- Test connectivity between New York and London PCs.
- Prove that traffic is protected, using the IKE security association and the encrypted/decrypted packet counters.

4. Tools and Technologies

Component	Details
Simulator	Cisco Packet Tracer (free; no separate IOS image or subscription needed)
Routers	2 x Cisco 2911, IOS 15.1(4)M4, Security Technology Package (securityk9) enabled
Switches	2 x Cisco 2960-24TT
End devices	10 x PC-PT (5 per office)
Routing	Static routes
VPN	IPsec (ESP) with IKE Phase 1 (ISAKMP), pre-shared key authentication
Verification	ping, show ip route, show crypto isakmp sa, show crypto ipsec sa

5. Theory Used

Static routing. An administrator enters each route by hand. NY-R1 is told that 192.168.20.0/24 is reached via 10.0.0.2, and LON-R1 is told that 192.168.10.0/24 is reached via 10.0.0.1. Static routes provide reachability only. They give no protection to the traffic.

IPsec. IPsec is a framework that provides confidentiality (encryption), integrity (tamper detection) and peer authentication at the IP layer. In this project the Encapsulating Security Payload (ESP) protocol is used in

tunnel mode, so the whole original packet, including the private source and destination addresses, is encrypted and wrapped in a new packet between the two router WAN addresses.

IKE Phase 1 (ISAKMP). The two routers authenticate each other with a pre-shared key and agree on a secure management channel using AES-256, SHA hashing and Diffie-Hellman group 5. When the state shows **QM_IDLE**, Phase 1 is established.

IKE Phase 2 (IPsec). The routers negotiate the IPsec security associations (SAs) that protect user data. The transform set *esp-aes esp-sha-hmac* specifies AES encryption with SHA-HMAC integrity.

Interesting traffic and crypto map. An extended ACL (110) selects which traffic is protected: here, traffic between 192.168.10.0/24 and 192.168.20.0/24. The crypto map ties together the peer, the transform set and the ACL, and is applied to the WAN interface.

6. Network Design

The diagram labels read NY-Router1 and LON-Router1. The routers' CLI hostnames are **NY-R1** and **LON-R1**, and these names are used in the rest of the report. The 192.168.10.1 and 192.168.20.1 labels beside the switches are the routers' LAN interface addresses (G0/0).

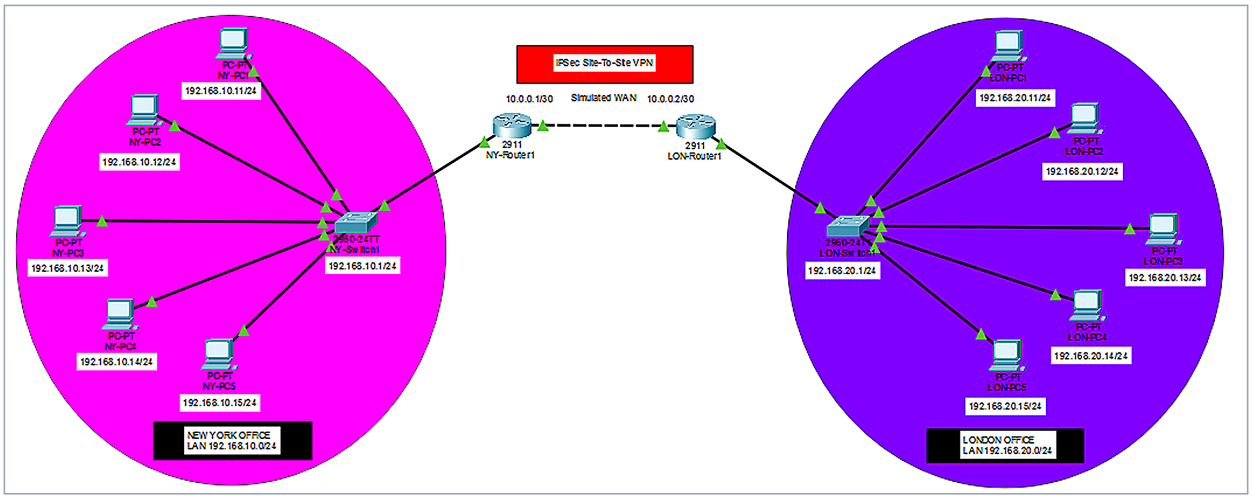


Figure 1: Network topology: two branch LANs joined by a simulated WAN carrying the IPsec tunnel.

Device	Interface	IP address	Subnet mask	Gateway
NY-R1	G0/0 (LAN)	192.168.10.1	255.255.255.0	-
NY-R1	G0/1 (WAN)	10.0.0.1	255.255.255.252	-
LON-R1	G0/0 (LAN)	192.168.20.1	255.255.255.0	-
LON-R1	G0/1 (WAN)	10.0.0.2	255.255.255.252	-
NY-PC1 to NY-PC5	Fa0	192.168.10.11 to .15	255.255.255.0	192.168.10.1
LON-PC1 to LON-PC5	Fa0	192.168.20.11 to .15	255.255.255.0	192.168.20.1

7. Implementation: Base Network

Step 1: Topology. Two routers, two switches and ten PCs were placed in Packet Tracer. PCs connect to the switches with straight-through cables, each switch connects to its router, and the two routers are joined by a cross-over cable on G0/1, which represents the WAN.

Step 2: PC addressing. Every PC was given a static address, mask and default gateway. Figures 2 and 3 show NY-PC1 and LON-PC1 as examples.



Figure 2: NY-PC1 IP configuration (192.168.10.11/24, gateway 192.168.10.1).

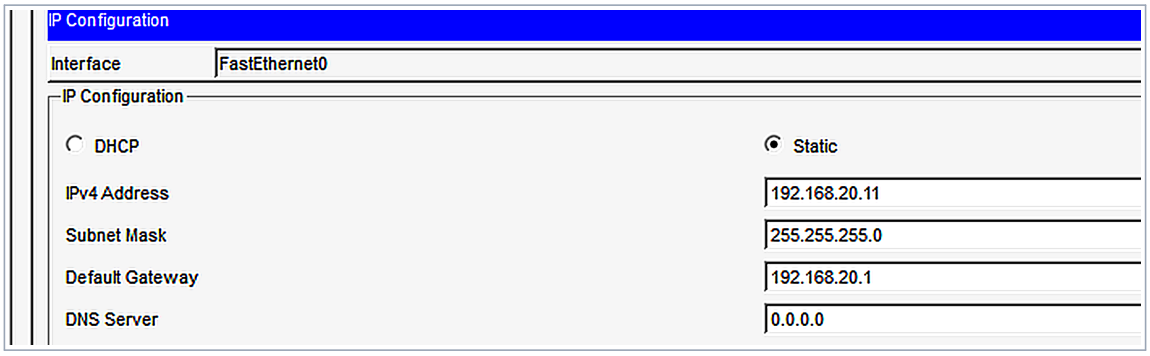


Figure 3: LON-PC1 IP configuration (192.168.20.11/24, gateway 192.168.20.1).

Step 3: Router interfaces and static routes. LAN and WAN interfaces were configured and enabled, and one static route was added on each router. Both routers show G0/0 and G0/1 as up/up.

```
NY-R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES NVRAM  up                    up
GigabitEthernet0/1     10.0.0.1        YES NVRAM  up                    up
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

Figure 4: NY-R1: show ip interface brief (G0/0 192.168.10.1, G0/1 10.0.0.1, both up/up).

```
LON-R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.20.1    YES NVRAM  up                    up
GigabitEthernet0/1     10.0.0.2        YES NVRAM  up                    up
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

Figure 5: LON-R1: show ip interface brief (G0/0 192.168.20.1, G0/1 10.0.0.2, both up/up).

Static route on NY-R1. The routing table contains the static route *S 192.168.20.0/24 [1/0] via 10.0.0.2*, which sends London-bound traffic to LON-R1. LON-R1 has the mirror route, *ip route 192.168.10.0 255.255.255.0 10.0.0.1*, which appears in its running configuration in Appendix B.

```
NY-R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
S    192.168.20.0/24 [1/0] via 10.0.0.2
```

Figure 6: NY-R1: show ip route showing the static route to the London LAN.

8. IPsec VPN Configuration

On the Cisco 2911 in Packet Tracer the *crypto* commands are hidden until the Security Technology Package is enabled with *license boot module c2900 technology-package securityk9* followed by a reload. Figure 7 confirms that **securityk9** is active.

```
NY-R1#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
*0         CISCO2911/K9          FTX15240F6F-


Technology Package License Information for Module:'c2900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
```

Figure 7: NY-R1: show version (excerpt) showing securityk9 enabled (Evaluation).

The VPN configuration on NY-R1 is explained below. LON-R1 uses the same settings with the peer address 10.0.0.1 and the mirror-image ACL.

Command	Purpose

`crypto isakmp policy 10`
`encr aes 256`
`hash sha`
`authentication pre-share`
`group 5`
	IKE Phase 1. Defines the secure management channel: AES-256 encryption, SHA hashing, pre-shared-key authentication and Diffie-Hellman group 5. (SHA is the IOS default, so *hash sha* does not appear in the running-config.)

`crypto isakmp key VPNsecret123 address 10.0.0.2`	The shared secret used to authenticate the peer at 10.0.0.2.
`crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac`	**IKE Phase 2.** Protects user data with ESP: AES encryption and SHA-HMAC integrity.
`access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255`	Defines the interesting traffic: only New York LAN to London LAN traffic is encrypted. London uses the reverse ACL.

`crypto map VPN-MAP 10 ipsec-isakmp`
`set peer 10.0.0.2`
`set transform-set VPN-SET`
`match address 110`
	Binds the peer, the transform set and the ACL into one policy.

`interface g0/1` `crypto map VPN-MAP`	Activates the policy on the WAN interface, so matching packets leaving toward London are encrypted.

```
NY-R1#
NY-R1#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
        Peer = 10.0.0.2
        Extended IP access list 110
            access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
        Current peer: 10.0.0.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                VPN-SET,
        }
        Interfaces using crypto map VPN-MAP:
                GigabitEthernet0/1

NY-R1#
```

Figure 8: NY-R1: show crypto map. The map uses peer 10.0.0.2, transform set VPN-SET and ACL 110, and is applied to GigabitEthernet0/1.

9. Testing and Results

Connectivity test. NY-PC1 pinged its gateway and then LON-PC1, and LON-PC1 pinged its gateway and then NY-PC1. All replies were received with 0% loss. The TTL of 126 on inter-office replies is the initial 128 minus two router hops, which shows that the packets crossed both routers.

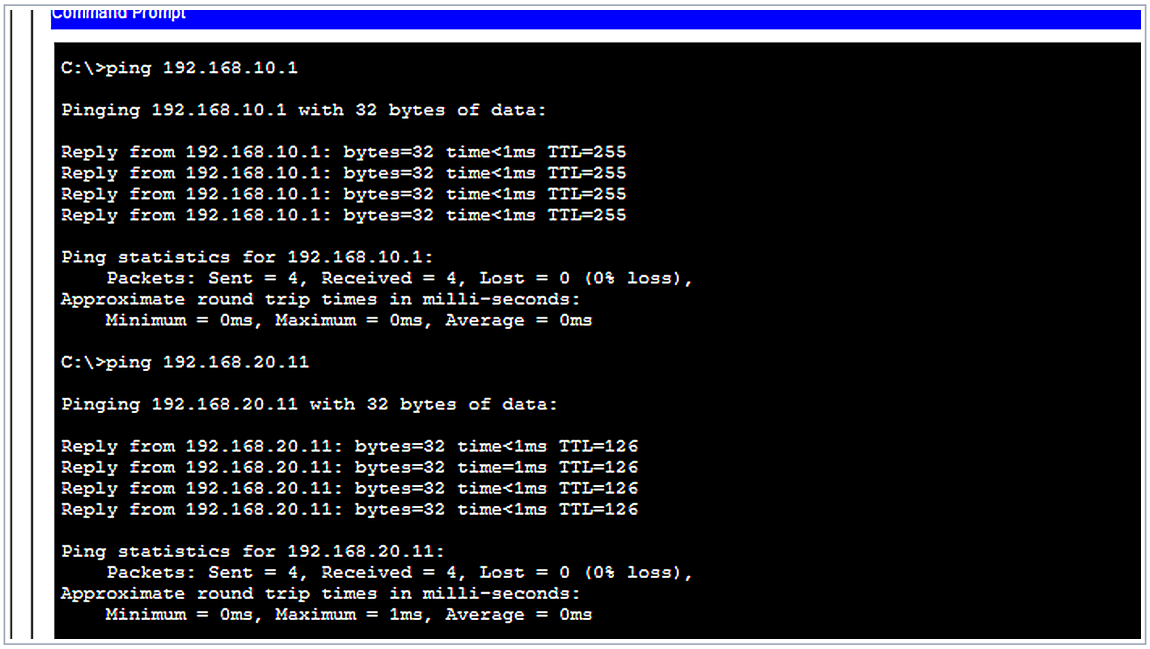


Figure 9: NY-PC1: ping to its gateway 192.168.10.1 and to LON-PC1 (192.168.20.11), 0% loss.

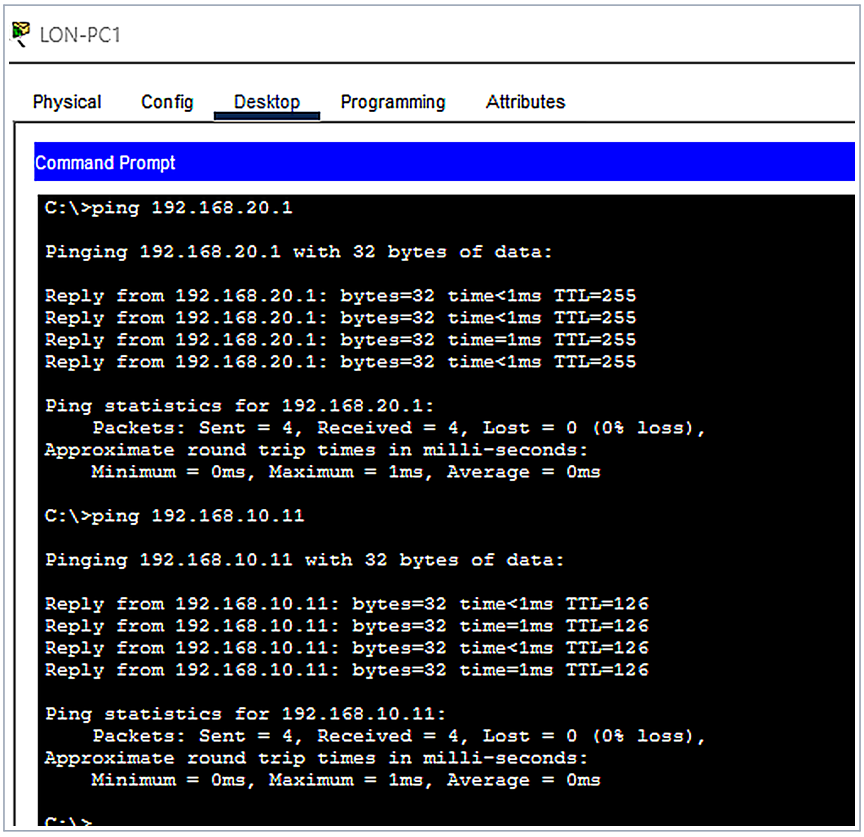


Figure 10: LON-PC1: ping to its gateway 192.168.20.1 and to NY-PC1 (192.168.10.11), 0% loss.

Why a ping alone is not proof of the VPN. The static routes give connectivity even with no VPN. To show the difference, the crypto map was temporarily removed from NY-R1's WAN interface and the ping was repeated. The ping still worked, but *show crypto ipsec sa* reported no security associations. Once the crypto map was re-applied, the same ping created the tunnel and the encrypted packet counters began to rise.

```
NY-R1#show crypto ipsec sa

No SAs found
NY-R1#
```

Figure 11: NY-R1 with the crypto map removed: show crypto ipsec sa reports "No SAs found" (nothing is encrypted).

10. VPN Verification

IKE Phase 1. Both routers show an ISAKMP security association in state **QM_IDLE** with status ACTIVE between 10.0.0.1 and 10.0.0.2. ACL 110 on NY-R1 also records matches, confirming that traffic is being selected for protection.

```
NY-R1#show access-lists
Extended IP access list 110
    10 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 (2 match(es))

NY-R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.0.0.2        10.0.0.1        QM_IDLE           1027    0 ACTIVE


IPv6 Crypto ISAKMP SA
```

Figure 12: NY-R1: show access-lists (ACL 110 matches) and show crypto isakmp sa (QM_IDLE, ACTIVE).

```
LON-R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.0.0.1        10.0.0.2        QM_IDLE           1039    0 ACTIVE


IPv6 Crypto ISAKMP SA
```

Figure 13: LON-R1: show crypto isakmp sa (QM_IDLE, ACTIVE).

IPsec security associations and counters. The highlighted counters show packets encrypted by one router and decrypted by the other. The SAs use *esp-aes esp-sha-hmac* in tunnel mode and have status ACTIVE. The peer values and ESP transform confirm that the traffic crosses the WAN as ESP, not as plain ICMP.

```
NY-R1#show crypto ipsec sa

interface: GigabitEthernet0/1
    Crypto map tag: VPN-MAP, local addr 10.0.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   current_peer 10.0.0.2 port 500
    PERMIT, flags={origin is acl,}
   #pkts encaps: 62, #pkts encrypt: 62, #pkts digest: 0
   #pkts decaps: 57, #pkts decrypt: 57, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 1, #recv errors 0

     local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
     path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
     current outbound spi: 0x6AF25721(1794266913)
```

Figure 14: NY-R1: show crypto ipsec sa. Local network 192.168.10.0/24, remote network 192.168.20.0/24, peer 10.0.0.2. Encaps/encrypt = 62 and decaps/decrypt = 57.

```
LON-R1#show crypto ipsec sa

interface: GigabitEthernet0/1
    Crypto map tag: VPN-MAP, local addr 10.0.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   current_peer 10.0.0.1 port 500
    PERMIT, flags={origin is acl,}
   #pkts encaps: 61, #pkts encrypt: 61, #pkts digest: 0
   #pkts decaps: 66, #pkts decrypt: 66, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.0.2, remote crypto endpt.:10.0.0.1
     path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
     current outbound spi: 0x0916C024(152485924)
```

Figure 15: LON-R1: show crypto ipsec sa. Local network 192.168.20.0/24, remote network 192.168.10.0/24, peer 10.0.0.1. Encaps/encrypt = 61 and decaps/decrypt = 66.

Note: the two outputs were captured at different moments (the remaining key lifetimes, 3367 s on NY-R1 and 3177 s on LON-R1, differ by about 190 seconds). For that reason the counters do not match exactly. Taken at the same instant, packets encrypted on one side equal packets decrypted on the other.

11. Summary of Results

Test	Expected result	Observed result
PC to gateway ping (both offices)	Replies	Success, 0% loss
Router interfaces	up/up	Success (NY-R1 and LON-R1)
Static routes	Remote LAN learned via the peer WAN address	Present on both routers
NY PC to London PC ping	Replies	Success, TTL 126
London PC to NY PC ping	Replies	Success, TTL 126
IKE Phase 1	QM_IDLE / ACTIVE	Success on both routers
IPsec SAs	Inbound and outbound ESP SAs active	Success (AES, SHA-HMAC, tunnel mode)
Encrypted traffic	Encrypt/decrypt counters above zero	Success (about 60 packets each way)
Without crypto map	No SAs	"No SAs found" (confirms VPN is what encrypts)

12. Limitations and Future Scope

- The WAN is a single direct link in a simulator, not the real internet. In a real network the tunnel would cross several provider routers, and a NAT or firewall device might need to allow IKE (UDP 500) and ESP.
- The pre-shared key and the IKEv1 crypto-map method are used for teaching. Production networks typically use IKEv2, stronger Diffie-Hellman groups, certificate-based authentication and long, unique keys.
- Routes are static. A dynamic routing protocol over GRE-over-IPsec or VTI tunnels would scale better, and a second tunnel could provide redundancy.
- The configuration could be extended with DHCP for the PCs, NAT for internet access, and ACLs to restrict which hosts may use the tunnel.

13. Conclusion

The project built two five-PC branch LANs, joined their edge routers over a simulated WAN, and configured static routes and an IKEv1/IPsec tunnel between them. Ping tests succeeded in both directions. The ISAKMP security association reached QM_IDLE, the IPsec security associations were active, and the encrypted and decrypted packet counters increased with traffic. When the crypto map was removed, the ping still worked but no SAs existed, which shows that the static routes provide reachability and IPsec provides the protection. The project demonstrates the skills a network engineer needs to link branch offices securely: addressing, routing, VPN configuration and verification.

Appendix A: NY-R1 Running Configuration (excerpt)

Blank separator lines and the spanning-tree section were removed to save space. An empty *router rip* process is present in this configuration but advertises no networks; all inter-office routing is static.

```
NY-R1#show running-config
Building configuration...
Current configuration : 1210 bytes

version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname NY-R1

ip cef
no ipv6 cef

license udi pid CISCO2911/K9 sn FTX15240F6F-
license boot module c2900 technology-package securityk9

crypto isakmp policy 10
 encr aes 256
 authentication pre-share
 group 5

crypto isakmp key VPNsecret123 address 10.0.0.2

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

crypto map VPN-MAP 10 ipsec-isakmp
 set peer 10.0.0.2
 set transform-set VPN-SET
 match address 110


interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 duplex auto
 speed auto

interface GigabitEthernet0/1
 ip address 10.0.0.1 255.255.255.252
 duplex auto
 speed auto
 crypto map VPN-MAP

interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
interface Vlan1
 no ip address
 shutdown

router rip

ip classless
ip route 192.168.20.0 255.255.255.0 10.0.0.2

ip flow-export version 9

access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255

line con 0

line aux 0

line vty 0 4
 login

end
```

Figure 16: NY-R1 running configuration: securityk9 license, IKE policy, key, transform set, crypto map, interfaces, static route and ACL 110.

Appendix B: LON-R1 Running Configuration (excerpt)

The configuration mirrors NY-R1: the peer address is 10.0.0.1, the static route points to the New York LAN, and ACL 110 is the reverse of the one on NY-R1.

```
LON-R1#show running-config
Building configuration...
Current configuration : 1198 bytes

version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption

hostname LON-R1

ip cef
no ipv6 cef

license udi pid CISCO2911/K9 sn FTX15246G2U-
license boot module c2900 technology-package securityk9

crypto isakmp policy 10
 encr aes 256
 authentication pre-share
 group 5

crypto isakmp key VPNsecret123 address 10.0.0.1

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac

crypto map VPN-MAP 10 ipsec-isakmp
 set peer 10.0.0.1
 set transform-set VPN-SET
 match address 110


interface GigabitEthernet0/0
 ip address 192.168.20.1 255.255.255.0
 duplex auto
 speed auto

interface GigabitEthernet0/1
 ip address 10.0.0.2 255.255.255.252
 duplex auto
 speed auto
 crypto map VPN-MAP

interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown

interface Vlan1
 no ip address
 shutdown

ip classless
ip route 192.168.10.0 255.255.255.0 10.0.0.1

ip flow-export version 9

access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255

line con 0

line aux 0

line vty 0 4
 login

end
```

Figure 17: LON-R1 running configuration: securityk9 license, IKE policy, key, transform set, crypto map, interfaces, static route and ACL 110.